MILITARY SPECIFICATION

RESIN, HEAT-RESISTANT, LAMINATING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of resin used in fabricating reinforced plastic structural parts for aircraft, missiles, and other applications in which extreme heat resistance is essential.

1.2 Classification. The resin shall be of the following type and classes (see 6.2):

Type I - Polyimide (PI)

Class 1 - 10 - 50 psi laminating pressure.
Class 2 - 50 - 300 psi laminating pressure.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-D-705	Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon)
PPP-D-729	Drums, Metal 55-Gallon (for Shipment of Noncorrosive Material)
PPP-P-704	Pails, Shipping, Steel (1 through 12 Gallon)

STANDARDS

Federal

Federal Test Method Standard No. 406 Plastics: Methods of Testing

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads (40 Inch x 48 Inch, 4-Way Partial and 4-Way Pallets)

(Copies of documents, other than specifications and standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Specifications and standards are available from the U. S. Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

D 635	Test for Flammability of Rigid Plastics Over 0.127 cm in Thickness.
D 638	Test for Tensile Properties of Plastics
D 695	Test for Compressive Properties of Rigid Plastics
D 790	Test for Flexural Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Ill. 60606.)

Code of Federal Regulations

49 CFR 171-178	Transportation of Explosives and Other Dangerous Articles by Land and Water.

(The Department of Transportation (DOT) Regulations are a part of the Code of Federal Regulations (latest cumulative pocket supplement) available from the Superintendent of Documents, Government Printing Office,

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Washington, D. C. 20402. Orders for the regulation should cite "Code of Federal Regulations, 49 CFR 171-178.".)

Manufacturing Chemists Association, Incorporated

Manual L-1 Guide to Precautionary Labeling of Hazardous Chemicals

(Application for copies should be addressed to the Manufacturing Chemists Association, Universal Building, 1825 Connecticut Avenue, Washington, D. C. 20009.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 4.3) a preproduction sample of the resin shall meet the requirements of this specification.

3.2 Material. The resin shall be heat-resistant, laminating and of polyimide (PI) type. The component materials of the resin shall be free of foreign matter and shall be such that the cured resin is not corrosive to metals.

3.2.1 Form. The resin shall be in liquid form suitable for impregnating, or may be already impregnated in an applicable reinforcement.

3.2.2 Coding. The manufacturer shall designate each resin composition by a code number which shall be used to identify the resin. The code number may include trade name, if desired. Any changes in the components or manufacturing procedures shall necessitate resubmission for acceptance, and a new code number shall be assigned.

3.2.2.1 Dry-layup materials, those using the same resin but having different properties because of different impregnating conditions, shall have different designations and shall be submitted independently for acceptance as different materials. The designation shall consist of the resin code number with a suffix indicating impregnating condition.

3.3 Limiting values. The manufacturer shall submit limiting values within which the specific gravity (4.7.3) and viscosity (4.7.4) of the liquid resin can be controlled. For preimpregnated material, limiting values shall be furnished for the volatile and dry resin content (4.7.5), flow (4.7.7) and gel time (4.7.8). These values are to be included in the instruction sheet (see 3.6). When approved, these values shall be

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considered requirements of this specification.

3.4 Mechanical and physical properties. The mechanical and physical properties of laminates of the resin shall conform to the values given in table I. These properties were developed using an aminosilane finish and avoiding content of less than 2 percent.

Table I. Mechanical and physical properties.

Property	Test 1/ Condition	Type I Requirement Class 1 & 2	Methods	
			ASTM	FTMS 406
Flexural strength (psi) ultimate	1 2 3	65,000 28,000 15,000	D 790	1031
Initial modulus of elasticity	1 2 3	3.3×10^6 2.5×10^6 1.2×10^6		
Tensile strength (psi)	1 2	50,000 20,000	D 638	1011
Compressive strength (psi)	1 2	55,000 25,000	D 695	1021
Interlaminar shear strength (psi)	1	2,000	-	1042
Flammability (in/min)	1	Self-extinguishing	D 635	2021

1/ Test conditions shall be as follows:

1. Ambient conditions (see 4.4.1)
2. @ 550°F after 1000 hr @ 550°F (±5°F forced air oven).
3. @ 700°F after 100 hr @ 700°F (±5°F forced air oven).

3.5 Electrical properties of the laminate. The electrical properties of a glass cloth base laminate, fabricated as specified in 4.3.1.2, shall conform to the values given in table II.

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Table II. Electrical requirements for glass cloth base plastic laminates at frequencies in the X-band range.

Type	Condition	Dielectric constant	Loss tangent
I	Standard	4.00 ± 0.15	0.007 ± 0.002

Notes: 1. The recommended x-band test frequency is 9,375 megacycles per second.
2. Values reported shall be the average of the determinations made on a minimum of three specimens.

3.6 Manufacturer's instruction sheet. The supplier shall provide an instruction sheet for each unit package of resin in the liquid or pre-impregnated form.

3.6.1 For liquid resin. The approved instruction sheet for the liquid resin shall contain the following information:

a. Type of the resin and the number of this specification.
b. Form of the resin and percent solids content (see 3.2.1).
c. Resin code number (see 3.2.2).
d. Limiting values required in 3.3.
e. Maximum usable storage life of the resin and recommended storage condition.
f. Recommended mixing and impregnating procedures, including recommended type, percentage, and manner of utilization of catalysts and retardants.
g. Recommended finishes to be used on glass fabric.
h. Maximum allowable shelf life at various temperatures of impregnated fabric before curing.
i. Range of time, temperature, and pressure cycles recommended to affect the cure for laminates.
j. Any other pertinent information, at the supplier's option, on storage and handling.

3.6.2 For preimpregnated material. The approved instruction sheet for the preimpregnated form, with description of the method of determining the properties, shall contain the following information:

a. Type of the resin and the number of this specification.

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b. Form of the resin (3.2.1).
c. Material designation (see 3.2.2).
d. Maximum allowable shelf life at various temperatures of impregnated fabric before curing.
e. Volatile content (average and tolerance) (see 3.3).
f. Dry resin content (average and tolerance) (see 3.3).
g. Flow (average and tolerance) (see 3.3).
h. Gel (average and tolerance) (see 3.3).
i. Range of time, temperature and pressure cycles recommended to effect the cure for laminates.
j. Any other pertinent information, at the supplier's option, on storage and handling.

3.7 Storage life. The storage life of the resin material shall be as recommended by the manufacturer.

3.8 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring agency.

3.9 Workmanship. The heat resistant laminating material shall be processed as required in accordance with the best practice for high quality resin.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the resin shall be classified as:

(a) Preproduction tests (4.3)
(b) Quality conformance test (4.5)

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4.3 Preproduction tests.

4.3.1 Preproduction sample. The preproduction sample shall consist of 1-gallon of resin (not required for dry lay-up) or sufficient impregnated fabric to fabricate a 2 x 2 foot x 1/8 inch thick laminate. Two copies of the instruction sheet specified in 3.6 shall be furnished with the sample.

4.3.1.1 Fabrication of laminate for physical test purposes. A glass fabric base plastic laminate panel shall be fabricated in the form of a flat sheet employing a parallel lay-up, using the resin to be tested and 10 plies of 181 glass fabric. The laminate shall be fully cured in accordance withthe resin manufacturer's instruction sheet. The laminate shall be essentially void-free and representative of the best workmanship and shall have a resin content by weight which will produce laminates with optimum properties. Specimens from this panel shall be tested to determine conformance with the requirements of this specification. Six specimens shall be used for each test.

4.3.1.2 Fabrication of laminate for electrical test purposes. A glass fabric base laminate shall be fabricated in the form of a flat sheet employing a parallel lay-up, using the resin to be tested and enough plies of fabric to give a sample 0.60 to 0.70 inch thick.

4.3.2 Identification. Preproduction test specimens shall be forwarded to the activity responsible for testing, as specified in the contract or order (see 6.2), plainly identified by securely attached durable tags marked with the following information:

a. Sample for preproduction test.
b. Specification number
c. Resin, or dry lay-up, Type and Class.
d. Manufacturer's name or trade name.
e. Manufacturer's designation and code number.

4.4 Test conditions.

4.4.1 Ambient conditions. Ambient conditions surrounding tests shall be 70° to 80°F and 30 to 60 percent relative humidity. If it is found that these conditions affect the test results significantly, then tests shall be conducted under standard conditions. Standard conditions shall be 73.4° ±2°F and 50 ±4 percent relative humidity. Specimens shall be tested after being exposed for four days to this temperature and humidity.

4.4.2 Exposure at elevated temperatures.

4.4.2.1 Exposure at 550 and 700°F. Specimens shall be exposed for the

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required time (see table I) at the applicable temperature $\pm 5^{\circ}F$ in the test chamber previously heated to the temperature of test and tested immediately at the same temperature.

4.5 Quality conformance test.

4.5.1 Lot size. All material offered for delivery at one time shall be considered a lot for the purpose of quality conformance testing.

4.5.2 Sampling for material conformance. Two quarts of resin or sufficient dry lay-up material to fabricate a 1 x 2 foot by 1/8 inch thick laminate shall be selected at random from each lot and shall be subjected to the tests specified in 4.7.4, 4.7.5, 4.7.6, and 4.7.8 for the resin, and in accordance with table I for the dry lay-up.

4.5.3 Sampling for filled container conformance. A random sample of filled resin containers shall be selected from each lot in accordance with MIL-STD-105, inspection level I, acceptable quality level of 2.5 percent defective, and shall be subjected to the examination specified in 4.6.2.

4.5.3.1 Any container in the sample having one or more defects or which is under required fill, shall be cause for rejection, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. (See 4.9)

4.6 Examinations.

4.6.1 Examination for preparation for delivery. An examination shall be made to determine that packaging, packing, and marking comply with specified contract or order requirements.

4.6.2 Examination of filled containers. Each unit sample selected as specified in 4.5.3 shall be visually examined for defects in accordance with MIL-STD-105 or table III.

4.7 Tests.

4.7.1 Mechanical and physical properties. The mechanical and physical properties of the laminate shall be determined in accordance with the methods specified in table I and as follows.

4.7.2 Electrical properties. The electrical properties of the laminate shall be determined in accordance with the method 4021 in Federal Test Method Standard No. 406.

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Table III
Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type not as specified, (oblong, square or round).
102	Gauge of steel not within requirements.
103	Size (capacity) not as specified.
104	Container not coated or coating nonconforming, (internal and external, as applicable).
105	Means of closure not as required; closure (cover) leaks, (container inverted); gasket, when applicable, missing, damaged or nonconforming.
106	Seam improperly formed, soldered or welded.
107	Seam leaks.
108	Not the required volume (fill).
109	Marking (as to content) not as specified, missing or illegible.
Minor:	
201	Not free from dents, scratches, burrs or sharp exposed edges.
202	Evidence of rust or corrosion.
203	Bails improperly attached or missing (when applicable).
204	Diameter of fail wire less than allowable minimum (when applicable).
205	Fail wire hand grip missing, nonconforming, not wood or metal as required (when applicable.

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4.7.3 Specific gravity. Specific gravity of the liquid resin shall be determined by weighing a specific volume of the resin to the nearest 0.001 gram and dividing the volume in cubic centimeters.

4.7.4 Viscosity. Viscosity of the liquid resin shall be determined by a calibrated McMichael viscosimeter, or equivalent. The test procedure shall be submitted by the manufacturer.

4.7.5 Volatile content and dry resin content. The test method shall be determined by the manufacturer (see 3.3, 3.6.1 and 3.6.2).

4.7.6 Solids content. The test method shall be determined by the manufacturer (see 3.6.1)

4.7.7 Flow of preimpregnated material (dry lay-up). The flow of preimpregnated material shall be determined in the following manner. Four 4-inch squares shall be cut from the impregnated fabric material. These specimens shall be weighed. Specimens of 4 plies shall be wrapped in cellophane and placed between platens of the flow meters before being pressed into a laminate under conditions specified by the manufacturer. After removing the specimen from the flow meter the cellophane, as well as all the flash resin that has flowed from the laminate, shall be removed from the 4 edges. The specimen shall then be reweighed. This weight shall be subtracted from the original weight, divided by the original weight, and multiplied by 100 to obtain the flow (see 3.6.2).

4.7.8 Gel time. The test method shall be determined by the manufacturer. (see 3.6.1 and 3.6.2)

4.8 Toxicological data and formulations. The supplier shall furnish the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

4.9 Rejection criteria. Failure of any lot sample to fully meet the requirements herein shall reject the lot represented. Disposition of inspection lots found unacceptable under initial conformance test shall be in accordance with MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The resin shall be furnished in 1-gallon cans, 5-gallon pails, 16 or 55-gallon drums, as specified (see 6.2).

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5.1.1.1 Cans. Cans shall conform to Type V, Class 2, round, of PPP-C-96. Cans shall be fabricated from tin plate having a minimum coating of 9.25 ounces per base box. Exterior plan B coating shall be used in accordance with PPP-C-96. Cans shall be provided with wire handles which shall be galvanized or protectively coated to resist corrosion.

5.1.1.2 Pails. The 5-gallon pails shall conform to Type II, Class 3 of PPP-P-704. Wire handles or bails shall be galvanized or protectively coated to resist corrosion.

5.1.1.3 Drums.

5.1.1.3.1 Sixteen-gallon drums. The 16-gallon drums shall conform to Type III of PPP-D-705.

5.1.1.3.2 Fifty-five gallon drums. The 55-gallon drums shall conform to Type III or Type IV of PPP-D-729.

5.1.2 Level C. Resin shall be packaged in accordance with the contractor's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cans. Cans shall be arranged and packed for shipment in accordance with the appendix to PPP-C-96.

5.2.1.2 Pails and drums. No further packing required. When specified (see 6.2), 5-gallon pails and 16-gallon drums shall be palletized in accordance with MIL-STD-147.

5.2.2 Level B.

5.2.2.1 Cans. Cans shall be arranged and packed for shipment in accordance with the appendix to PPP-C-96.

5.2.2.2 Pails and drums. No further packing required. When specified (see 6.2), 5-gallon pails and 16-gallon drums shall be palletized in accordance with MIL-STD-147.

5.2.3 Level C. Material shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination.

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Shipping containers or method of packing shall comply to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking for shipment. In addition to any special marking required by the contract or order or herein, interior packages, exterior shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129. The nomenclature shall include:

RESIN, HEAT-RESISTANT, LAMINATING
MIL-R-

Resin _____ Type_____, Class_____
Batch Number _____

5.3.1 Hazardous chemicals. All packaged units of hazardous chemicals shall have affixed thereto such warning labels and markings as may be required by the Department of Transportation (DOT) Regulations, 49 CFR 171-178, and the Manufacturing Chemists Association Manual L-1.

5.3.2 Warning markings. All packaged units shall be marked with the following:

WARNING: USE WITH ADEQUATE VENTILATION - FLAMMABLE

The solvents contained in this compound may have flash points well below prevailing room temperatures. Do not heat by, or use near an open flame. Heat preferably by means of a steam coil or other nonflame-producing heat source.

6. NOTES

6.1 Intended use. Polyimide or laminating resin may be used in fabricating laminates for high temperature structural parts and other applications.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.
b. Type and class.
c. Where the qualification samples should be sent and the activity responsible for testing (see 4.3.2).
d. Type and size of container required (see 5.1.1).
e. Levels of packaging and packing required (see 5.1 and 5.2).
f. If palletization is required (see 5.2.1.2 and 5.2.2.2).
g. Frequency of periodic lot check testing.

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Custodians:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Army - AV, EL
Navy - AS
Air Force - 84

User activity:
Army - MI
Navy - EC, MC, OS, SH

Preparing activity:
Air Force - 11

Project No.: 9330-0493

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. *GIVE PARAGRAPH NUMBER AND WORDING.*

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. *IS THE SPECIFICATION RESTRICTIVE?*

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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